Martín Vicente Maza Delgado

Valdivia, Chile • martin.maza@alumnos.uach.cl • github.com/mittods • linkedin.com/in/martinmazadelgado

Education

Universidad Austral de Chile (UACh) Valdivia, Chile
Civil Engineering in Computer Science (*Ingeniero Civil en Informática*) Expected: June 2027

- 5.5-year engineering degree, ∼330 SCT credits (286 completed, no failed courses). **Ranked 5th of 81** in my cohort.
- Licentiate in Engineering Sciences (*Licenciado en Ciencias de la Ingeniería*), obtained after the 8th semester (240 SCT).
- Relevant courses: Software Engineering; Software Engineering Methods and Models; Software Architecture; Requirements Engineering; Distributed Systems; High-Performance Computing; Operating Systems; Networks; Artificial Intelligence.

Master in Informatics (*Magíster en Informática*) Expected: June 2028

- Started through the joint undergraduate–master's track: master's courses taken during the final undergraduate year.

Research

Fondecyt Project 11260438, UACh March 2026 – Present
Research collaborator (PI: Prof. Valeria Henríquez)

- Project: *University Software Engineering Education in the Era of Generative AI: Diagnosis, Intervention and Evaluation in Chile.*
- Designed multi-agent LLM pipelines for systematic literature reviews (search, screening, evidence extraction, synthesis), with an independent verification step that checks each claim against its source.

Undergraduate thesis (article-based research project) 2026 – 2027

- *A multidimensional model for diagnosing curricular obsolescence in software engineering courses in the face of generative AI.* Includes an automated pipeline that evaluates real exams with LLM APIs, and expert validation (Delphi).

Publications (under review)

- I. Cabrera, M. Maza, V. Henríquez, J. P. Salazar-Fernandez. "Occupationally Transversal Skills in SFIA 9: Implications for Competency-Based Computing Engineering Education in the GenAI Era." Submitted to IEEE EDUNINE 2027.
- M. Alvarado, C. Vergara, M. Maza, J. P. Salazar-Fernandez, V. Henríquez. "Teaching Software Engineering with Generative AI: Connecting SWEBOK, Cognitive Processes, and Didactic Techniques." Submitted to IEEE EDUNINE 2027.

Technical Experience

Honner Chile – Full-Stack Developer and Product Architecture Remote, Aug 2024 – Present

- Designed and built a real-time monitoring platform for manufacturing companies: machinery instrumented with sensors, PLCs and edge computers (NUCs), with a distributed web architecture; exploring agent-based integrity monitoring of the platform.
- Integrated on-premise infrastructure with cloud services; production setup (domain, SSL, availability); CI/CD and containerisation.

Sophia Language Technologies – Software Developer and Architecture Support Valdivia, Mar 2026 – Present

- AI conversational assistant with RAG for clinical treatment follow-up: backend integration and patient-facing interfaces.
- Proposed architecture improvements for service communication and availability; set up repositories and CI/CD deployment.

Valdiv-IA Project, UACh – Scrum Master and DevOps/Full-Stack Developer Valdivia, Aug 2025 – Dec 2025

- City-scale conversational assistant: RAG over local data combining open models self-hosted with Ollama and the OpenAI API, Redis vector store; Docker deployment on an institutional server.

Automotriz Salfa Sur – Internship, Infrastructure and Development Puerto Montt, Dec 2025 – Feb 2026

- Internal RAG assistant with locally served models (Ollama, Python, Redis); IT inventory system and internal APIs (Laravel, PostgreSQL).
- Prototyped a failover VPN router (Raspberry Pi, StrongSwan, iptables) to keep network access during internet outages.

Teaching and Service

Teaching Assistant, Institute of Informatics, UACh 2024 – Present

- Programming; Object-Oriented Programming; Software Construction Workshop; Software Engineering Workshop; Software Engineering; Algorithms.

IEEE Student Branch UACh – Secretary (2026 – Present), Webmaster (2025 – 2026); competitive programming workshops; working towards a joint Computer Society / Computational Intelligence Society chapter.

Competitions and Skills

Competitions: ICPC Latin America South Regional (2025); IEEEXtreme 19.0 (2025); Chilean Programming Tournament (2025, 2026).

Languages: Spanish (native); English (advanced; UACh academic writing in English course, 2024).

Technical: Python, C++, TypeScript/JavaScript, PHP, SQL; Linux, Docker, Git, CI/CD; PostgreSQL, Redis; React, Node.js, Laravel; RAG with self-hosted open models (Ollama) and commercial LLM APIs (OpenAI).